

Practice Sheet 1: Answers

Equation from Intercepts - 1 mark for the gradient, 1 for the full equation

Section A

A1 $c = 6$. $m = -6 / 3 = -2$ So $y = -2x + 6$ 2 marks

A2 $c = 4$. $m = -4 / -2 = 2$ So $y = 2x + 4$ 2 marks

A3 $c = -8$. $m = 8 / 4 = 2$ So $y = 2x - 8$ 2 marks

A4 $c = 5$. $m = -5 / 5 = -1$ So $y = -x + 5$ 2 marks

A5 $c = -9$. $m = 9 / -3 = -3$ So $y = -3x - 9$ 2 marks

A6 $c = -8$. $m = 8 / 2 = 4$ So $y = 4x - 8$ 2 marks

Section B

B1 $c = 4$. $m = -4 / 6 = -\frac{2}{3}$ So $y = -\frac{2}{3}x + 4$ 2 marks

B2 $c = 2$. $m = -2 / -4 = \frac{1}{2}$ So $y = \frac{1}{2}x + 2$ 2 marks

B3 $c = -4$. $m = 4 / 10 = \frac{2}{5}$ So $y = \frac{2}{5}x - 4$ 2 marks

B4 $c = -9$. $m = 9 / -6 = -\frac{3}{2}$ So $y = -\frac{3}{2}x - 9$ 2 marks

B5 $c = 6$. $m = -6 / 9 = -\frac{2}{3}$ So $y = -\frac{2}{3}x + 6$
x 3: $3y = -2x + 18 \rightarrow$ $2x + 3y = 18$ 2 marks

B6 $c = 6$. $m = -6 / -8 = \frac{3}{4}$ So $y = \frac{3}{4}x + 6$
x 4: $4y = 3x + 24 \rightarrow$ $3x - 4y = -24$ 2 marks

Quick check: put the x-intercept into your equation. You should get $y = 0$.

Common slips: the sign of m (does the line go up or down?), and m upside down (it is up / along, not along / up).